LAB Assignment-9

(Mix Concepts)

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Subject : Unstructured Database Lab

Section : E

COLLECTION 1: **members**

```
{
  member_id: Number,
  name: String,
  plan: String,
  city: String,
  fee: Number
}
```

Table

member_id	name	plan	city	fee
1	Aarav	Gold	Delhi	5000
2	Meera	Silver	Mumbai	3000
3	Rohan	Gold	Bangalore	5500
4	Sneha	Platinum	Delhi	7000
5	Kunal	Silver	Chennai	3200

COLLECTION 2: **sessions**

```
{
  session_id: Number,
```

member_id: Number,

trainer: String,

duration: Number,

calories: Number

}

Table

session_id	member_id	trainer	duration	calories
101	1	Raj	60	400
102	2	Simran	45	300
103	3	Raj	70	500
104	4	Aman	80	600
105	5	Raj	50	350

QUESTIONS

```
mongosh mongodb://127.0.0.
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Soma Chatterjee>mongosh
Current Mongosh Log ID: 69d3530b9af7f1bb4e7c2906
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0
mongosh 2.8.2 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-04-04T20:26:52.211+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use gymDB_Soma_Chatterjee_CSF_013
switched to db gymDB_Soma_Chatterjee_CSF_013
gymDB_Soma_Chatterjee_CSF_013> db.createCollection("members")
{ ok: 1 }
gymDB_Soma_Chatterjee_CSF_013> db.createCollection("sessions")
{ ok: 1 }
gymDB_Soma_Chatterjee_CSF_013> db.members.insertMany([
|   {member_id:1, name:"Aarav", plan:"Gold", city:"Delhi", fee:5000},
|   {member_id:2, name:"Meera", plan:"Silver", city:"Mumbai", fee:3000},
|   {member_id:3, name:"Rohan", plan:"Gold", city:"Bangalore", fee:5500},
|   {member_id:4, name:"Sneha", plan:"Platinum", city:"Delhi", fee:7000},
|   {member_id:5, name:"Kunal", plan:"Silver", city:"Chennai", fee:3200}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d354029af7f1bb4e7c2907'),
    '1': ObjectId('69d354029af7f1bb4e7c2908'),
    '2': ObjectId('69d354029af7f1bb4e7c2909'),
    '3': ObjectId('69d354029af7f1bb4e7c290a'),
    '4': ObjectId('69d354029af7f1bb4e7c290b')
  }
}
```

```
gymDB_Soma_Chatterjee_CSF_013> db.sessions.insertMany([
|   {session_id:101, member_id:1, trainer:"Raj", duration:60, calories:400},
|   {session_id:102, member_id:2, trainer:"Simran", duration:45, calories:300},
|   {session_id:103, member_id:3, trainer:"Raj", duration:70, calories:500},
|   {session_id:104, member_id:4, trainer:"Aman", duration:80, calories:600},
|   {session_id:105, member_id:5, trainer:"Raj", duration:50, calories:350}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d3553d9af7f1bb4e7c290c'),
    '1': ObjectId('69d3553d9af7f1bb4e7c290d'),
    '2': ObjectId('69d3553d9af7f1bb4e7c290e'),
    '3': ObjectId('69d3553d9af7f1bb4e7c290f'),
    '4': ObjectId('69d3553d9af7f1bb4e7c2910')
  }
}
```

Q1. Display all members

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.members.find()
[
  {
    _id: ObjectId('69d354029af7f1bb4e7c2907'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2908'),
    member_id: 2,
    name: 'Meera',
    plan: 'Silver',
    city: 'Mumbai',
    fee: 3000
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2909'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 5500
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c290a'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c290b'),
    member_id: 5,
    name: 'Kunal',
    plan: 'Silver',
    city: 'Chennai',
    fee: 3200
```

Q2. Show name and plan only

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.members.find({}, {name:1, plan:1, _id:0})
[
  { name: 'Aarav', plan: 'Gold' },
  { name: 'Meera', plan: 'Silver' },
  { name: 'Rohan', plan: 'Gold' },
  { name: 'Sneha', plan: 'Platinum' },
  { name: 'Kunal', plan: 'Silver' }
]
```

Q3. Find members from Delhi

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.members.find({city:"Delhi"})
[
  {
    _id: ObjectId('69d354029af7f1bb4e7c2907'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c290a'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  }
]
```

Q4. Find sessions with duration > 60

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.sessions.find({duration:{$gt:60}})
[
  {
    _id: ObjectId('69d3553d9af7f1bb4e7c290e'),
    session_id: 103,
    member_id: 3,
    trainer: 'Raj',
    duration: 70,
    calories: 500
  },
  {
    _id: ObjectId('69d3553d9af7f1bb4e7c290f'),
    session_id: 104,
    member_id: 4,
    trainer: 'Aman',
    duration: 80,
    calories: 600
  }
]
```

Q5. Sort members by fee descending

```
mongosh mongodb://127.0.0.
gymDB_Soma_Chatterjee_CSF_013> db.members.find().sort({fee:-1})
[
  {
    _id: ObjectId('69d354029af7f1bb4e7c290a'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2909'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 5500
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2907'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c290b'),
    member_id: 5,
    name: 'Kunal',
    plan: 'Silver',
    city: 'Chennai',
    fee: 3200
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2908'),
    member_id: 2,
    name: 'Meera',
    plan: 'Silver',
    city: 'Mumbai',
    fee: 3000
```

Q6. Update fee +500 for Gold members

```
mongosh mongodb://127.0.0.
gymDB_Soma_Chatterjee_CSF_013> db.members.updateMany(
|    {plan:"Gold"},
|    {$inc:{fee:500}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Q7. Delete members with fee < 3500

```
mongosh mongodb://127.0.0.
gymDB_Soma_Chatterjee_CSF_013> db.members.deleteMany({fee:{$lt:3500}})
{ acknowledged: true, deletedCount: 2 }
```

Q8. Insert a new member

```
gymDB_Soma_Chatterjee_CSF_013> db.members.insertOne({
|   member_id:6,
|   name:"Rahul",
|   plan:"Gold",
|   city:"Kolkata",
|   fee:6000
| })
{
  acknowledged: true,
  insertedId: ObjectId('69d355c29af7f1bb4e7c2911')
}
```

Q9. Average fee per plan

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.members.aggregate([
|   {$group:{_id:"$plan", avgFee:{$avg:"$fee"}}}
| ])
[
  { _id: 'Platinum', avgFee: 7000 },
  { _id: 'Gold', avgFee: 5833.333333333333 }
]
```

Q10. Total calories burned per trainer

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.sessions.aggregate([
|   {$group:{_id:"$trainer", totalCalories:{$sum:"$calories"}}}
| ])
[
  { _id: 'Simran', totalCalories: 300 },
  { _id: 'Raj', totalCalories: 1250 },
  { _id: 'Aman', totalCalories: 600 }
]
```

Q11. Maximum fee per city

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.members.aggregate([
|   {$group:{_id:"$city", maxFee:{$max:"$fee"}}}
| ])
[
  { _id: 'Bangalore', maxFee: 6000 },
  { _id: 'Delhi', maxFee: 7000 },
  { _id: 'Kolkata', maxFee: 6000 }
]
```

Q12. Count members per plan

```
mongosh mongodb://127.0.0.
gymDB_Soma_Chatterjee_CSF_013> db.members.aggregate([
|    {$group:{_id:"$plan", count:{$sum:1}}}
| ])
[ { _id: 'Platinum', count: 1 }, { _id: 'Gold', count: 3 } ]
```

Q13. Trainers with avg calories > 400

```
mongosh mongodb://127.0.0.
gymDB_Soma_Chatterjee_CSF_013> db.sessions.aggregate([
|    {$group:{_id:"$trainer", avgCalories:{$avg:"$calories"}}},
|    {$match:{avgCalories:{$gt:400}}}
| ])
[
  { _id: 'Aman', avgCalories: 600 },
  { _id: 'Raj', avgCalories: 416.6666666666667 }
]
```

Q14. Total duration per member_id

```
mongosh mongodb://127.0.0.
gymDB_Soma_Chatterjee_CSF_013> db.sessions.aggregate([
|    {$group:{_id:"$member_id", totalDuration:{$sum:"$duration"}}}
| ])
[
  { _id: 3, totalDuration: 70 },
  { _id: 1, totalDuration: 60 },
  { _id: 2, totalDuration: 45 },
  { _id: 5, totalDuration: 50 },
  { _id: 4, totalDuration: 80 }
]
```

Q15. Join members with sessions

```
mongosh mongodb://127.0.0.

gymDB_Soma_Chatterjee_CSF_013> db.members.aggregate([
|    {
|      $lookup:{
|        from:"sessions",
|        localField:"member_id",
|        foreignField:"member_id",
|        as:"session_details"
|      }
|    }
| ])
[
  {
    _id: ObjectId('69d354029af7f1bb4e7c2907'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500,
    session_details: [
      {
        _id: ObjectId('69d3553d9af7f1bb4e7c290c'),
        session_id: 101,
        member_id: 1,
        trainer: 'Raj',
        duration: 60,
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2909'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    session_details: [
      {
        _id: ObjectId('69d3553d9af7f1bb4e7c290e'),
        session_id: 103,
        member_id: 3,
```

```
mongosh mongodb://127.0.0.
    city: 'Bangalore',
    fee: 6000,
    session_details: [
      {
        _id: ObjectId('69d3553d9af7f1bb4e7c290e'),
        session_id: 103,
        member_id: 3,
        trainer: 'Raj',
        duration: 70,
        calories: 500
      }
    ]
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c290a'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000,
    session_details: [
      {
        _id: ObjectId('69d3553d9af7f1bb4e7c290f'),
        session_id: 104,
        member_id: 4,
        trainer: 'Aman',
        duration: 80,
        calories: 600
      }
    ]
  },
  {
    _id: ObjectId('69d355c29af7f1bb4e7c2911'),
    member_id: 6,
    name: 'Rahul',
    plan: 'Gold',
    city: 'Kolkata',
    fee: 6000,
    session_details: []
  }
]
```

Q16. Show members who have sessions

mongosh mongodb://127.0.0.

```
gymDB_Soma_Chatterjee_CSF_013> db.members.aggregate([
|   {
|     $lookup:{
|       from:"sessions",
|       localField:"member_id",
|       foreignField:"member_id",
|       as:"sessions"
|     }
|   },
|   {
|     $match:{"sessions.0":{$exists:true}}
|   }
| ])
[
  {
    _id: ObjectId('69d354029af7f1bb4e7c2907'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500,
    sessions: [
      {
        _id: ObjectId('69d3553d9af7f1bb4e7c290c'),
        session_id: 101,
        member_id: 1,
        trainer: 'Raj',
        duration: 60,
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2909'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    sessions: [
      {
```

```
mongosh mongodb://127.0.0.
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2909'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    sessions: [
      {
        _id: ObjectId('69d3553d9af7f1bb4e7c290e'),
        session_id: 103,
        member_id: 3,
        trainer: 'Raj',
        duration: 70,
        calories: 500
      }
    ]
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c290a'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000,
    sessions: [
      {
        _id: ObjectId('69d3553d9af7f1bb4e7c290f'),
        session_id: 104,
        member_id: 4,
        trainer: 'Aman',
        duration: 80,
        calories: 600
      }
    ]
  }
]
```

Q17. Create index on city

```
gymDB_Soma_Chatterjee_CSF_013> db.members.createIndex({city:1})
city_1
```

Q19. Create compound index on plan and fee

```
gymDB_Soma_Chatterjee_CSF_013> db.members.createIndex({plan:1, fee:-1})
plan_1_fee_-1
```

Q20. Display 2nd and 3rd highest fee members

```
gymDB_Soma_Chatterjee_CSF_013> db.members.find()
|    .sort({fee:-1})
|    .skip(1)
|    .limit(2)
[
  {
    _id: ObjectId('69d355c29af7f1bb4e7c2911'),
    member_id: 6,
    name: 'Rahul',
    plan: 'Gold',
    city: 'Kolkata',
    fee: 6000
  },
  {
    _id: ObjectId('69d354029af7f1bb4e7c2909'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000
  }
]
```